A $\sigma_g > 0$: the rewards differ

one prompt x, drawn from the step's batch B_t

↓

G completions sampled i.i.d.; together they form one group g

y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8
1	0	1	1	0	1	0	0

scalar rewards from one shared verifier

↓

$$\mu_g = \frac{1}{G}\sum_{i=1}^{G} r_{g,i} = 0.500$$

$$\sigma_g = \sqrt{\frac{1}{G-1}\sum_{i=1}^{G}(r_{g,i}-\mu_g)^2} = 0.535$$

$$A_{g,i} = (r_{g,i} - \mu_g)/(\sigma_g + \varepsilon)$$

$\varepsilon = 10^{-6}$; unbiased sample variance (ddof = 1)

↓

Reward contrast is present: some $A_{g,i} \neq 0$. This does not guarantee a nonzero total parameter update.

This group is counted in GU = 1 − ZVF, not in ZVF.

B $\sigma_g = 0$: the rewards are all identical

one prompt x, drawn from the step's batch B_t

↓

G completions sampled i.i.d.; together they form one group g

y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8
0	0	0	0	0	0	0	0

the same verifier returns the same value for all G

↓

$$\mu_g = \frac{1}{G}\sum_{i=1}^{G} r_{g,i} = 0.000$$

$$\sigma_g = \sqrt{\frac{1}{G-1}\sum_{i=1}^{G}(r_{g,i}-\mu_g)^2} = 0.000$$

$$A_{g,i} = (r_{g,i} - \mu_g)/(\sigma_g + \varepsilon)$$

every reward identical $\Rightarrow \sigma_g = 0$

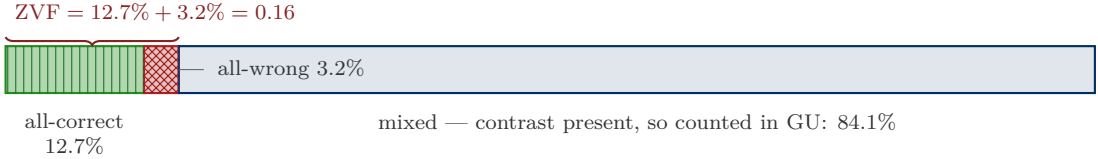
↓

Advantage is identically zero: every $A_{g,i} = 0$, so the group has no reward-relative term. Other loss gradients can remain.

It is counted in ZVF; here the cause is all-wrong, but all-correct is identical.

Definitions. $\mathrm{ZVF}_{0,t} = \dfrac{1}{|B_t|}\displaystyle\sum_{g\in B_t} \mathbf{1}[r_{g,1} = r_{g,2} = \cdots = r_{g,G}] = \dfrac{\#\text{ groups whose } G \text{ rewards are identical}}{\#\text{ groups in the step's batch } B_t}$

$\mathrm{GU}_{0,t} = 1 - \mathrm{ZVF}_{0,t}$ (gradient utilisation: the share of groups that carry within-group contrast)

Exact ties require zero variance. The report's thresholded ZVF_ε instead counts sample variance $\widehat{\mathrm{Var}} \leq 10^{-6}$, including near-ties. They agree for these binary examples, but differ after micro-jitter.

Measured composition. Qwen3-8B on a held-out GSM8K slice via the Tinker API, $G = 8$, $T = 1.0$, 200 problems × 3 seeds: mean ZVF = 0.16, hence GU = 0.84.